

PHISHING EMAIL ANALYSIS & IOC EXTRACTION

SOC Analyst Lab Project

Lab Date: 31 March 2026

Report Date: 31 March 2026

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Environment: Windows 11

Tools Used: Windows 11, Notepad, MxToolbox, VirusTotal

1. INTRODUCTION

1.1 Project Objective

The main objective of this project is to analyze a phishing email and extract Indicators of Compromise (IOCs). This project demonstrates how SOC analysts investigate suspicious emails, identify malicious indicators, and document findings.

1.2 Scope

This project covers:

- Creating a sample phishing email (.eml file)
- Extracting raw email headers using Notepad
- Analyzing headers with MxToolbox
- Identifying suspicious sender domain and IP
- Scanning malicious IP with VirusTotal
- Documenting IOCs

1.3 Tools Used

Tool	Purpose
Windows 11	Primary analysis machine
Notepad	View raw email headers
MxToolbox	Email header analysis
VirusTotal	IP reputation check

2. LAB SETUP

2.1 Lab Environment

Machine	Operating System	Role
Analyst	Windows 11	Email analysis, header extraction, IOC scanning

2.2 Environment Description

This project is executed on a single Windows 11 machine. No network connectivity is required beyond accessing online tools (MxToolbox, VirusTotal). A sample phishing email was created locally in .eml format.

3. CREATE PHISHING EMAIL

3.1 Email Content Creation

Content:

From: security@paypal-support.com

To: victim@company.com

Subject: URGENT: Your PayPal Account Has Been Limited

Dear Customer,

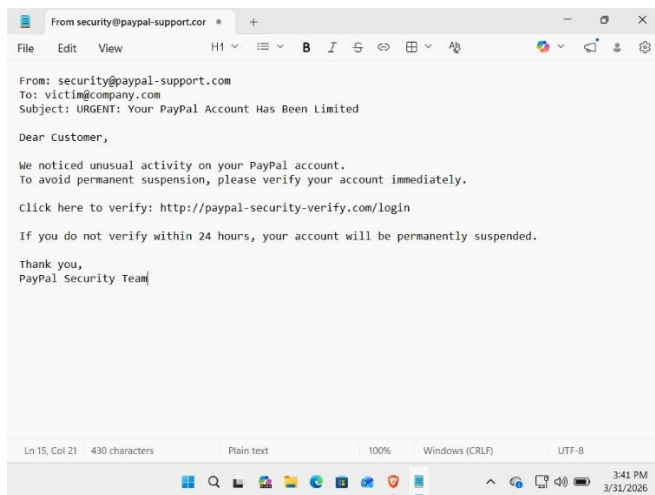
We noticed unusual activity on your PayPal account.
To avoid permanent suspension, please verify your account immediately.

Click here to verify: <http://paypal-security-verify.com/login>

If you do not verify within 24 hours, your account will be permanently suspended.

Thank you,
PayPal Security Team

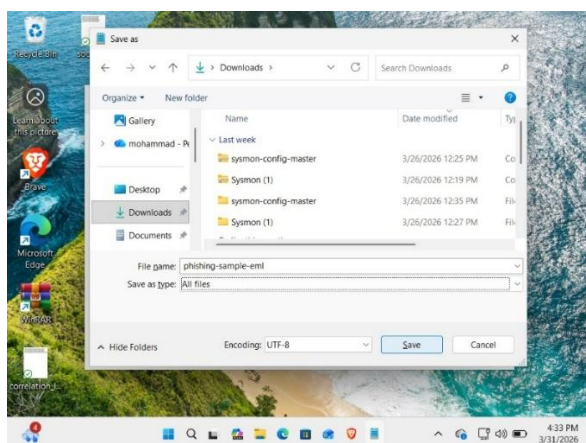
Screenshot 1 – Email Content



3.2 Save as .eml File

- Save file as phishing-sample.eml
- **Save as type:** All Files (.)

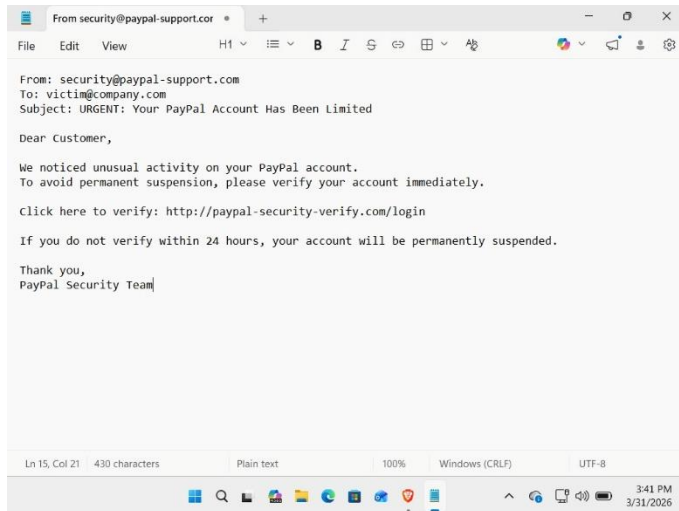
Screenshot 2 – Email Saved



3.3 Verify Email

- Open .eml file in Notepad or default email client

Screenshot 3 – Email Opened

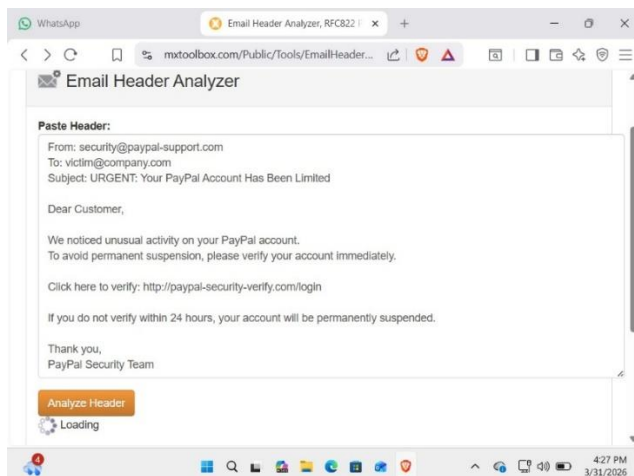


4. EMAIL HEADER ANALYSIS

4.1 Extract Raw Headers

- Open phishing-sample.eml in **Notepad**
- Copy entire content (headers + body)

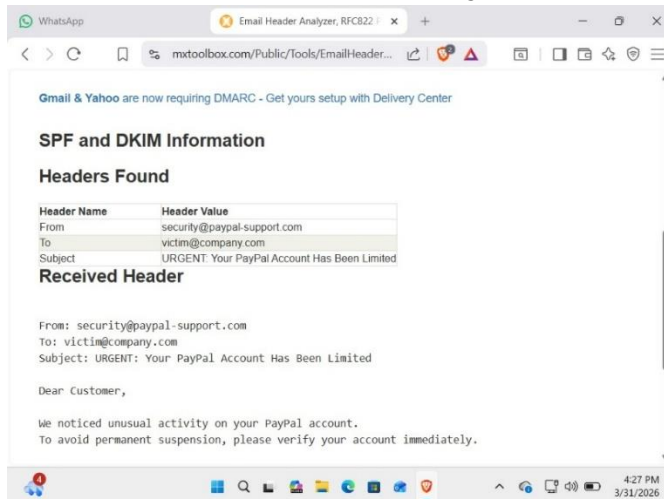
Screenshot 4 – Email Headers Pasted in MxToolbox



4.2 Header Analysis Using MxToolbox

1. Go to: <https://mxtoolbox.com/EmailHeaders.aspx>
2. Paste copied headers
3. Click Analyze Headers

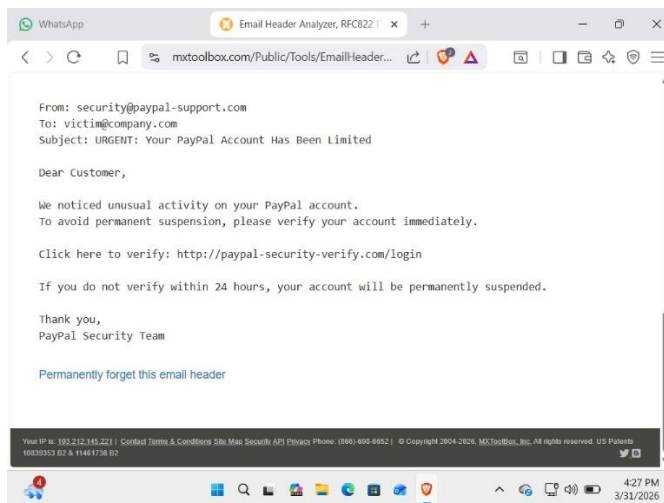
Screenshot 5A – MxToolbox Analysis



4.3 Suspicious IP Identification

IP found in headers: 103.212.145.221

Screenshot 5B – IP in Headers



4.4 IP Reputation Check Using VirusTotal

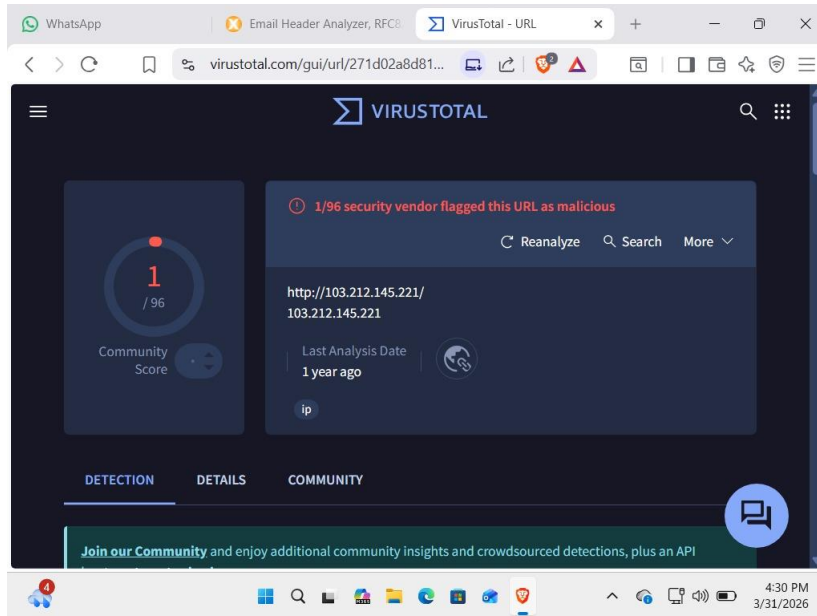
1. Go to: <https://www.virustotal.com/gui/home/url>
2. Search IP: 103.212.145.221

Result:

- 1/96 security vendors flagged this IP as malicious

- Although only 1/96 vendors flagged this IP, combined with the suspicious sender domain, fake PayPal branding, and phishing URL — this IP is still considered an IOC in the context of this phishing email

Screenshot 6 – VirusTotal IP Analysis



5. INDICATORS OF COMPROMISE (IOCs)

5.1 IOCs Identified

IOC NO.	Type	Value	Description
IOC 01	Email Address	security@paypal-support.com	Fake sender domain
IOC 02	Domain	paypal-support.com	Suspicious domain (not official PayPal)
IOC 03	URL	http://paypal-security-verify.com/login	Phishing link
IOC 04	Subject	URGENT: Your PayPal Account Has Been Limited	Urgency-based social engineering
IOC 05	IP Address	103.212.145.221	Malicious IP (1/96 flagged by VirusTotal)
IOC 06	Content	"Verify your account immediately"	Pressure tactic

5.2 IOC Severity Levels

IOC NO.	Severity	Justification
IOC 01	Critical	Fake sender — impersonation attempt
IOC 02	Critical	Suspicious domain — typosquatting

IOC 03	Critical	Direct phishing link
IOC 04	High	Urgency tactic to bypass user caution
IOC 05	Critical	Malicious IP flagged by VirusTotal
IOC 06	Medium	Common phishing pressure tactic

6. CONCLUSION

6.1 Project Summary

Field	Value
Project Name	Phishing Email Analysis + IOC Extraction
Objective	Analyze a phishing email and extract Indicators of Compromise (IOCs)
Tools Used	Windows 11, Notepad, MxToolbox, VirusTotal
Environment	Windows 11 (Local Machine)

6.2 Key Achievements

- Created a sample phishing email (.eml file)
- Extracted raw email headers using Notepad
- Analyzed headers using MxToolbox
- Identified suspicious sender domain and IP
- Scanned malicious IP with VirusTotal (1/96 flagged)
- Documented 6 IOCs with severity levels

6.3 Challenges Faced & Solutions

Challenge	Solution
MxToolbox showed limited results	Still extracted key IOCs (From, Subject, IP)
IP found in headers	Verified with VirusTotal (1/96 flagged)

6.4 Learning Outcomes

Skill Area	What I Learned
Email Analysis	How to extract and analyze email headers
Phishing Indicators	Identifying fake sender domains, urgency tactics
IOC Extraction	Documenting indicators from suspicious emails
Threat Intelligence	Using VirusTotal for IP reputation checks

6.5 Future Scope

- Analyze email attachments with sandbox (Any.Run)
- Check domain reputation using WHOIS lookup
- Create automated phishing detection rules in SIEM
- Simulate phishing campaigns for security awareness

7. GITHUB REPOSITORY

<https://github.com/Mubashir-soc-01/SOC-Project>

8. LINKEDIN PROFILE

<https://www.linkedin.com/in/mohammad-mubashir-b729643b7/>

THANK YOU

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